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How Far Can the Gemma 4 Speech Encoder Go Without Changing Its Weights? Saudi and Gulf Arabic Speech Recognition with a Frozen and an Adapter-Augmented Encoder

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Abstract

We study the 305M-parameter audio encoder shared by Gemma 4 and EmbeddingGemma 2 (identical except for its final projection) for Saudi and Gulf Arabic speech recognition, without changing any of its weights. A frozen system trains only a 22.6M-parameter CTC reader over a mix of encoder layers; an adapted system adds 3.1M low-rank adapter parameters, a 41.5M-parameter hybrid CTC/attention reader and joint decoding with a sub-word language model. Against Cohere Transcribe Arabic (2.07B parameters), re-run on identical audio, the frozen system trails in word error rate (WER) but is statistically indistinguishable in character error rate (CER) on five of six human-labelled conditions, while running 3.4 times faster on CPU with greedy decoding. The adapted system is statistically indistinguishable in WER on a public Saudi customer-service call (19.20% against 19.27%) and on clean Emirati speech, is significantly better on telephone-coded Emirati speech, has a significantly lower CER on all six conditions of these sets, and beats a 0.6B recogniser fine-tuned on SADA in WER on all six; it still trails Cohere in WER on long-form Saudi speech. On a blind, human-corrected public call-centre sample it ranks 4 of 15 systems, behind three commercial APIs (significantly only one) and ahead of the other self-hostable systems by point estimate. An optimised joint decoder runs it at 139 times real time on one GPU. We also show that references made by correcting machine drafts favour the drafting system. Code and public-set results are released.

1 Introduction

Arabic speech recognition is hardest where it matters most commercially: dialectal speech over the telephone. Saudi and Gulf call centres mix regional dialects, code-switching, and spoken digit strings such as identity and account numbers. All of this arrives on 8 kHz narrow-band audio. Public benchmarks confirm the difficulty. On the Saudi SADA test set, the best open systems in the Open Universal Arabic ASR Leaderboard still reach word error rates (WER) above 40% [1]. A dedicated benchmark of Arabic call-domain speech reports WERs near 50%, and this holds even for large commercial recognisers [2].

The usual remedies are to fine-tune a large pretrained recogniser [3–6] or to train a dedicated Arabic model on many hours of labelled speech [7, 8]. Both demand substantial compute for every new domain. A cheaper option keeps a strong speech encoder frozen and trains only a small task head on top of it. Probing benchmarks such as SUPERB and ML-SUPERB [9, 10] show that frozen self-supervised encoders retain enough phonetic information for recognition. Layer-wise analyses go further and reveal that this information peaks in intermediate layers rather than at the output [11, 12].

Open multimodal language models now ship with their own speech encoders. Gemma 4 [13] attaches a 305M-parameter Conformer audio encoder, derived from the Universal Speech Model

design [14], to a text decoder, and the same encoder has since been reused elsewhere: Google states that EmbeddingGemma 2, a multimodal embedding model, "shares its text tokenizer and audio encoder" with Gemma 4 [15, 16]. We confirmed this directly. Of the 752 audio-encoder tensors in the released EmbeddingGemma 2 checkpoint, 751 are bit-identical to those of Gemma 4 E2B [17]; the only difference is the final projection into each model's text embedding space, which a recogniser does not use. The encoder is therefore a component that developers may already be running, inside an assistant model or an embedding service, and it is released under the Apache-2.0 licence. Whether it can be reused for Arabic transcription without fine-tuning has, to our knowledge, not been studied.

This paper asks a narrow question. How far can this speech encoder go on Saudi and Gulf Arabic, including telephone speech, when its weights stay unchanged or are adapted only through a small number of added parameters? We make five contributions.

1. A *frozen* recipe: the unchanged Gemma 4 audio encoder (as distributed with EmbeddingGemma 2), a learned mix over its hidden layers, a 22.6M-parameter transformer reader trained with connectionist temporal classification (CTC) [18], and a word 4-gram language model in beam search [19, 20].

2. An *adapted* recipe that keeps the encoder's weights frozen but adds 3.1M low-rank adapter parameters [21] to six middle layers, with a 41.5M-parameter hybrid CTC/attention reader [22], self-conditioned CTC [23] and joint CTC/attention decoding [24] with a sub-word language model.

3. An evaluation on SADA, three further human-labelled Saudi and Gulf test sets, a blind human-corrected public call-centre sample with fifteen recognisers including twelve commercial services, and real call-centre recordings, against open recognisers re-run on identical audio: Cohere Transcribe Arabic (2.07B parameters) [25] and Nemotron 3.5 ASR (0.6B) [26], zero-shot and fine-tuned.

4. We present a faster implementation of joint CTC/attention decoding with a language model. On one GPU, it raises throughput from 1.7 to 139 times real time. Accuracy is essentially unchanged. We also measure speed and memory on GPU and CPU under a single protocol. Paired bootstrap intervals [27] and N-best oracle analyses then separate model errors from ranking errors.

5. Call-centre references produced by correcting machine drafts are mostly the unedited drafts. Such references favour the system that produced them, so we report results split by reference origin.

The results depend on the measure. The frozen system stays behind the 2B-parameter recogniser in WER on every test set. Its character error rate, however, is statistically indistinguishable from that recogniser's on five of the six human-labelled conditions, and it runs several times faster on CPU. The adapted system is statistically indistinguishable from the 2B recogniser in WER on the public customer-service call and on clean Emirati speech, and better on telephone-coded Emirati speech. On the further human-labelled sets it is significantly lower in CER across all six conditions. In WER, however, it remains behind on long-form SaudiTalk speech. Section 2 reviews related work, Section 3 describes the systems, Section 4 the data and protocol, Section 5 the results, Section 6 discusses limitations, and Section 7 concludes.

2 Related Work

2.1 Dialectal and Telephone Arabic ASR

Arabic recognition research has moved from Modern Standard Arabic broadcast speech towards dialects. SADA [28] provides several hundred hours of Saudi television speech with human

transcripts, and it has become the standard Saudi test bed. Fine-tuning results on its clean test subset are reported in [29]. Casablanca [30] covers eight Arabic varieties, including Emirati and Yemeni, with human transcripts and code-switching annotations. The Open Universal Arabic ASR Leaderboard [1] evaluates open models on several of these sets under a common text normalisation. Large-scale weakly supervised training on Arabic is studied in [7], while ArTST [8] pre-trains a joint Arabic speech–text model. For the call domain, [2] released a benchmark of Arabic call-centre style speech, yet its audio is 16 kHz. No public Saudi corpus of real 8 kHz call-centre speech of useful size exists, only small free samples, so we combine real call recordings, which we cannot release, with public human-labelled sets and with telephone-coded versions of them.

2.2 Frozen Encoders and Layer Analysis

Self-supervised encoders such as wav2vec 2.0 [5] and w2v-BERT [31, 32] are commonly fine-tuned end to end. SUPERB [9] and ML-SUPERB [10] instead evaluate them frozen, feeding a learned weighted sum over layers into a light downstream model. This protocol is the basis of our reader. Frozen encoders of supervised recognisers are also strong general features. Whisper-AT [33], for example, tags audio events from the frozen Whisper encoder. Pasad et al. [11, 12] showed that phonetic and word information is strongest in middle layers and weaker at the top of pretrained speech models.

2.3 Speech Encoders in Multimodal and Embedding Models

Recent multimodal language models attach a speech encoder to a text decoder. SLAM-ASR [34] showed that a frozen encoder, a small projector and a language model can give strong English recognition. The Gemma 4 report [13] describes an audio encoder of about 305M parameters. That encoder uses two down-sampling convolutions and 12 Conformer layers [35], producing one frame every 40 ms. A separate line of work trains speech encoders for cross-modal or semantic alignment, including SAMU-XLSR [36], SONAR [37], SpeechCLIP [38] and CLAP [39]. Those models are optimised for utterance-level representations rather than transcription. The encoder studied here belongs to the first group, because it is the encoder of an assistant model reused unchanged in an embedding model. We are not aware of a published Arabic recognition study of it.

2.4 Reader Architectures and Decoding

CTC [18] gives a *n*on-autoregressive output layer. Hybrid CTC/attention models [22], intermediate CTC losses [40] and self-conditioned CTC [23] relax the conditional independence of CTC. Joint CTC/attention decoding [24] scores attention hypotheses with CTC prefix probabilities. Shallow fusion adds an external language model to sequence-to-sequence search [41]. External n-gram models remain a cheap way to add word-level context to CTC output [19]. Our frozen system keeps the reader deliberately simple, so that the measurement isolates the encoder. The adapted system then combines these established components.

3 Method

Figure 1 gives an overview of both systems.

3.1 Frozen Encoder

The model's own feature extractor converts audio at 16 kHz into log-mel features. The audio encoder, loaded from the EmbeddingGemma 2 checkpoint, runs in bfloat16 with all weights

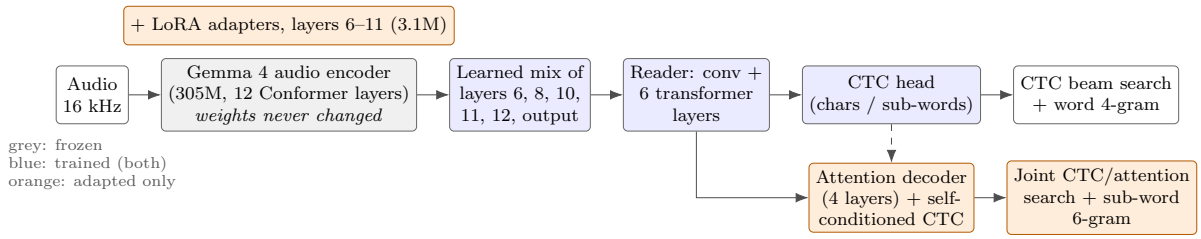

Figure 1: The two systems. The Gemma 4 audio encoder is never changed (grey). Both systems train a layer mix, a reader and a CTC head (blue); the adapted system adds low-rank adapters inside the encoder, an attention decoder with self-conditioned CTC and joint CTC/attention search (orange).

frozen. It has 12 Conformer-style layers of width 1024, followed by an output projection to 1536 dimensions, and it emits 25 frames per second. Its released configuration uses chunked local self-attention with no right context, so the encoder itself is causal. The reader described next is, however, not causal. We take the hidden states of layers 6, 8, 10, 11 and 12, together with the projected output, which gives six representations in total. A preliminary probe guided this subset, and the other layers were not tried in the mix. Telephone audio is up-sampled to 16 kHz before feature extraction.

3.2 Reader

Each selected representation h_ℓ is normalised and projected to 512 dimensions. The projections are then combined with softmax-normalised weights:

$$x_t = \sum_\ell \frac{\exp(a_\ell)}{\sum_k \exp(a_k)} W_\ell \,\mathrm{LN}(h_{\ell,t}), \tag{1}$$

where the a_ℓ are learned scalars initialised at zero. The mixed sequence passes through a residual depth-wise convolution (kernel 5), six pre-norm bidirectional transformer layers (512 dimensions, 8 heads, feed-forward size 2048, dropout 0.1) and a linear CTC output layer over 68 symbols: the Arabic letters left after normalisation, digits, the word boundary and the CTC blank. The reader has 22.6M parameters. During training, two random time spans per utterance, each up to 5% of its length, are zeroed in the mixed features, in the spirit of SpecAugment [42].

3.3 Training

We train the reader with the CTC loss. We use AdamW with a learning rate of 3×10^{-4} and weight decay 0.01, under a one-cycle schedule. Batches hold up to 6,000 encoder frames. Encoder features are computed on the fly, so the telephone simulation below can be applied afresh in each epoch. We select the model by validation character error rate on held-out calls.

3.4 Telephone Simulation

To expose the reader to narrow-band speech, wideband training audio is degraded on the fly. The audio is first down-sampled to 8 kHz, then encoded and decoded with G.711 μ-law [43], and finally up-sampled back to 16 kHz. The telephone versions of the test sets are produced by this same deterministic procedure.

3.5 Decoding

We report greedy CTC decoding and beam search with a word 4-gram language model [19] through pyctcdecode [20] (beam 64). The language model is trained on the training-call tran-

scripts and the SADA training transcripts, after the same normalisation as the scoring. Its weight and word-insertion bonus are tuned on the validation calls only, and the selected values are 0.5 and 2.0. Test data are never used for this tuning.

3.6 Adapted System

The adapted system keeps every encoder weight frozen but inserts low-rank adapters [21] (rank 16, scaling 32/16). They are placed first in the query, key, value and output projections of encoder layers 6 to 10 and later also in the feed-forward and convolution modules of layers 6 to 11, adding 3.1M trainable parameters in total. Its output units are 1,024 SentencePiece sub-words [44] instead of characters. The reader keeps the layer mix, convolution and six transformer layers of the frozen system and adds a CTC head and a four-layer autoregressive transformer decoder (512 dimensions, 8 heads) with gated sinusoidal positions; the reader has 41.5M parameters. Training proceeds in five warm-started stages: (1) sub-word CTC with the attention adapters; (2) the hybrid reader, initialised from stage 1, trained with CTC and attention losses for 6 epochs; (3) the decoder alone on the frozen reader, with label smoothing of 0.05 on content tokens, none on the end-of-sequence token and a weight of 1.5 on it, so that the decoder learns to stop; (4) all reader and adapter parameters jointly for 3 epochs with loss weights 0.3 (CTC), 0.5 (attention) and 0.2 (self-conditioned CTC [23], a shared head after reader layers 2 and 4 whose posteriors are fed back through a zero-initialised projection); and (5) the same joint training for 4 further epochs after adding zero-initialised adapters to the feed-forward and convolution modules, with 80 h of SADA per epoch. Every warm-started run first re-scores the validation set and must reproduce the source model's CTC result within one point. Each machine-labelled call window is weighted by the agreement of its two labels: windows whose labels differ by at most 25% CER get weight 1, by at most 35% weight 0.5, by at most 45% weight 0.25, and the rest are dropped; SADA utterances marked as unclear are removed.

The adapted system is decoded in two ways. The first, CTC beam search, uses a word 4-gram (weight 0.30, word bonus 1.50). The second, joint CTC/attention beam search [24] (CTC weight 0.4), adds a 6-gram over the model's own sub-words by shallow fusion [41] (weight 0.15, length bonus 1.00). Both language models are trained on the training-call transcripts, SADA training transcripts and MASC training transcripts [45], with all validation and test calls excluded. All decoding weights, and the choice between two language-model variants, were selected on the validation calls only. That selection followed stage 4 with a beam of 10, where the chosen variant scored WER 29.72% on validation against 30.05% for the alternative. The choices were then kept unchanged for stage 5. Test results use a beam of 16 and its top hypothesis.

4 Experimental Setup

4.1 Training Data

Call-centre speech The in-domain data are real Saudi call-centre recordings, taken as single-party legs of 8 kHz telephone calls and stored at 16 kHz. We cut them into windows of up to 30 s. Training labels are machine-generated: each window carries one transcript from Cohere Transcribe Arabic and one from a commercial cloud recogniser, and in each epoch one of the two is sampled per window. Calls reserved for validation and testing are kept out of both training and the language-model text. After filtering, 6,050 training windows with 12,089 labels remain. The validation set holds 295 windows drawn from 147 calls, and because these validation labels are also machine transcripts, validation scores serve only to select models.

Voicebot calls The adapted system additionally uses 1.63 h of real Saudi voicebot telephone calls (67 calls, 2,011 voice-activity segments), repeated four times per epoch. Their labels are

machine transcripts aligned to Cohere segment hypotheses. Segments whose machine transcript and Cohere hypothesis differ by more than 60% WER are dropped.

SADA The SADA training set [28] supplies human-labelled Saudi speech: 77,514 utterances (121.6 h), of which 60 h are sampled in each epoch. In the *telephone* variant, each SADA utterance is degraded as described above, and in the *clean* variant it is used as is.

4.2 Systems

We train three readers. The *calls-only* reader is trained on the call-centre windows for 20 epochs, and the best validation epoch falls at 9. Both the *telephone-SADA* reader and the *clean-SADA* reader begin from the calls-only reader and continue for 6 further epochs. Their training data consist of the calls plus SADA, and SADA is degraded to telephone quality in one case but kept clean in the other. For both, the best validation epoch is 3. We trained each configuration once. The adapted system of Section 3.6 starts from the same data, with the additions described above.

Two open recognisers serve as baselines. Both were run on exactly the same audio. They were scored with the same normalisation.

- **Cohere Transcribe Arabic** (07-2026, 2.07B parameters, Apache-2.0) [25, 46], a dedicated Arabic encoder–decoder recogniser. We use greedy decoding on voice-activity segments of at most 25 s, with a repetition penalty of 1.2 and no repeated 4-grams to prevent the looping that the model otherwise shows on silence.
- **Nemotron 3.5 ASR streaming 0.6B** [26], a cache-aware streaming FastConformer transducer. We use it zero-shot and after full fine-tuning on SADA only, in two stages of 4,000 steps each (learning rate 10^{-5}, cosine schedule): first on clean SADA, then on about 50 h of SADA coded with GSM to simulate telephony. It never saw the call-centre data. Decoding uses beam search of width 8, with the attention-context setting of 56 frames of left and 13 of right context.

Cohere Transcribe Arabic supplied one of the two machine labels for the call-centre training data, so the frozen-encoder system is partly distilled from it. The fine-tuned Nemotron model was trained on SADA alone.

4.3 Test Sets

- **SADA test** set: 4,971 human-labelled utterances of Saudi television speech [28], in clean and telephone-coded form.
- The **Saudi customer-service sample** is the free portion of a commercial corpus [47]. It consists of one 13-minute call leg, cut at its human time stamps into 48 turns. It is the closest public match to our domain. Still, it comes from a single call and a single speaker.
- **Casablanca, Emirati**: the 813 human-labelled utterances of the Emirati test split [30].
- **SaudiTalk** comprises 133 human-verified recordings in Hijazi, Ha'il and Southern dialects [48], amounting to about 6 h in total. Transcripts cover whole clips and carry no time stamps, so all systems decode voice-activity segments of at most 25 s, and the segment outputs are then joined per clip.
- The blind call-centre sample contains 92 clips, totalling 8.3 minutes and 854 reference words. The clips come from two one-sided mobile recordings in a public Saudi call-centre dialogue sample [49]. Clips were cut at voice-activity boundaries. Each clip received a draft transcript from one of three systems, assigned at random in equal shares and hidden from

the annotator. The author, a native speaker, then corrected each draft into a verbatim reference. Each system is scored only on clips its own draft did not produce. Pairs of systems are compared only on clips drafted by neither of them.

- The call-centre set contains 72 windows drawn from 63 held-out calls in the in-house recordings. The windows were deliberately selected to over-represent spoken numbers (29 windows) and windows on which the two machine transcripts disagreed most. Consequently, they are harder than typical call audio.

Every test set except the in-house calls is also scored after telephone coding.

How the call-centre references were made An annotator listened to each window in a correction tool prefilled with the longer of the two machine transcripts, and was asked to make it verbatim. In 63 of the 72 windows the draft was accepted unchanged: 30 Cohere drafts and 33 drafts from the commercial cloud recogniser. Only 9 were edited. The references are therefore mostly machine text checked by a single listener, and they favour the drafting systems. Cohere's own draft scores 30.39% WER against them, whereas the same model re-run on identical audio scores 37.62%. We treat this set as secondary and report it split by reference origin.

4.4 Metrics and Statistics

Text is normalised identically for every system: Unicode NFKC, removal of diacritics and tatweel, unification of alef, alef maqsura, taa marbuta and hamza carriers, collapse of letters repeated three or more times, removal of Arabic punctuation (comma, semicolon, question mark and similar marks), and removal of all characters other than Arabic letters, digits and spaces. Digits and spelled-out numbers are not converted into each other, so a number written in words by one system and in digits by the reference counts as an error. Our main WER and character error rate (CER, with spaces removed) are computed per utterance and averaged over utterances, which weights every utterance equally; we also report corpus-level WER (total errors over total reference words). CER ignores word boundaries, while WER counts a wrongly split or merged word as an error. Differences between systems are tested with a paired bootstrap over utterances (10,000 resamples) [27]; on the call-centre set we also resample whole calls, since several windows can come from one call. We report the bootstrap mean difference and its 95% percentile interval; these means can differ from the difference of the two printed scores in the last digit. References written mainly in Latin script (Arabic letters fewer than twice the Latin letters), which are translations rather than transcripts, are excluded; this removes one SADA utterance and none elsewhere. Validation figures quoted for model development were computed during development with an earlier normaliser that kept Arabic punctuation; every test figure in this paper uses the final normaliser, applied to saved hypotheses of every system.

5 Results

5.1 SADA

Table 1 reports the SADA test set. Adding SADA to training lowers WER with the language model from 59.84% to 48.55% on clean audio, and from 63.55% to 51.58% on telephone audio. The two SADA variants differ by less than one WER point. Training on telephone-coded SADA is 0.67 points better on telephone audio, while training on clean SADA is 0.19 points better on clean audio. In CER, telephone training helps more on telephone audio (26.85% against 28.15%). The simulated coding is therefore useful mainly at the character level.

Table 2 compares all systems on SADA test under the same normaliser. For context only, the leaderboard [1] reports 44.52% WER on SADA test for an NVIDIA Conformer-CTC model

Table 1: SADA test set (4,971 utterances): WER and CER (%) on the original 16 kHz audio and after simulated telephone coding (8 kHz).

	Clean 16 kHz			Telephone 8 kHz		
	WER		CER	WER		CER
Reader	greedy	+LM	+LM	greedy	+LM	+LM
Calls only	71.06	59.84	32.44	74.16	63.55	36.12
+ SADA, telephone-coded	59.98	48.55	24.16	63.33	51.58	26.85
+ SADA, clean	60.06	48.36	24.41	63.45	52.25	28.15

Table 2: SADA test (4,971 utterances), WER / CER (%) of every system under the same normaliser. Our systems use CTC search with the word 4-gram.

System	Clean	Telephone
Frozen (telephone-SADA reader)	48.55 / 24.16	51.58 / 26.85
Adapted	40.38 / 21.31	42.51 / 23.34
Nemotron 0.6B, fine-tuned on SADA	46.50 / 21.21	48.22 / 22.62
Cohere Transcribe Arabic 2B	46.85 / 27.59	47.88 / 29.00

with a language model, and 55.96% for Whisper large-v3 [3]. Cohere reports 37.47% for its model [46]. These figures use a different normalisation and corpus-level aggregation, so they are not directly comparable.

On SADA test the adapted system has the lowest WER of all systems: 40.38% on clean audio, against 46.85% for Cohere and 46.50% for Nemotron. The adapted system also has the lowest WER on telephone-coded audio. This comparison favours the systems trained on SADA. Our readers and the fine-tuned Nemotron model saw the SADA training set, whereas we do not know whether Cohere did. We therefore treat SADA as an in-domain benchmark rather than a fair test of generalisation. Nemotron has a slightly lower CER than the adapted system on this set (21.21% against 21.31% on clean audio).

5.2 Further Human-Labelled Test Sets

Table 3 compares the telephone-SADA reader with Cohere Transcribe Arabic on the three further sets, and Table 4 gives the paired differences.

Cohere is better in WER on every set, and every WER interval excludes zero: for example by 6.28 points on Emirati speech and by 10.35 on the customer-service sample. Corpus-level WER gives the same ordering. At the character level the picture differs. On Casablanca the CER gap is 0.91 points (interval -0.52 to 2.30), and on SaudiTalk it is 0.09 (interval -1.58 to 1.56). Both intervals include zero, and so does the customer-service call's (1.85 points, interval -0.44 to 4.00): in CER the frozen system is statistically indistinguishable from Cohere on every condition in Table 4 except telephone-coded SaudiTalk, where it is 2.96 points worse. The frozen encoder therefore recovers most of the letters but fewer whole words. Two factors likely contribute: CER ignores word boundaries, which the 4-gram model only partly restores, and Cohere's decoder carries a much stronger implicit language model than our 4-gram.

The language model lowers WER on every set but raises CER on every one of them. Without it, our reader's CER on Casablanca is already below Cohere's, at 19.93% against 20.35% on clean audio and 23.36% against 24.38% on telephone audio. On SaudiTalk it is 28.61% against Cohere's 29.51% on clean audio. The 4-gram model thus repairs word boundaries but sometimes replaces a correctly spelled dialect word with a more frequent one. A user who needs letters rather than words is better served by greedy decoding.

Table 3: Additional Saudi and Gulf test sets: WER / CER (%) of Cohere Transcribe Arabic and of the frozen-encoder system (telephone-SADA reader) with and without the language model.

Test set	n	Cohere	Ours + LM	Ours greedy
Saudi customer-service calls (clean)	48	19.27 / 10.05	29.62 / 11.91	34.50 / 11.07
Saudi customer-service calls (telephone)	48	20.53 / 10.92	30.63 / 12.72	37.04 / 12.13
Casablanca, Emirati (clean)	813	48.35 / 20.35	54.63 / 21.26	62.79 / 19.93
Casablanca, Emirati (telephone)	813	53.15 / 24.38	58.82 / 25.38	67.24 / 23.36
SaudiTalk long-form (clean)	133	56.75 / 29.51	65.29 / 29.60	70.17 / 28.61
SaudiTalk long-form (telephone)	133	55.80 / 28.40	66.56 / 31.36	72.46 / 30.47

Table 4: Paired difference, frozen-encoder system (+LM) minus Cohere, in percentage points, with 95% bootstrap intervals over utterances (10,000 resamples); the differences are bootstrap means of per-utterance differences. Negative values favour the frozen-encoder system. The last column gives corpus-level WER (total errors over total reference words).

Test set	n	ΔWER	ΔCER	Corpus WER (Cohere / ours)
Saudi customer-service calls (clean)	48	+10.35 [+7.16, +13.50]	+1.85 [-0.44, +4.00]	20.77 / 30.12
Saudi customer-service calls (telephone)	48	+10.10 [+7.49, +12.79]	+1.81 [-0.06, +3.65]	21.87 / 31.91
Casablanca, Emirati (clean)	813	+6.28 [+4.52, +8.05]	+0.91 [-0.52, +2.30]	47.52 / 53.53
Casablanca, Emirati (telephone)	813	+5.67 [+3.74, +7.57]	+1.00 [-0.60, +2.59]	51.75 / 57.32
SaudiTalk long-form (clean)	133	+8.54 [+6.89, +10.20]	+0.09 [-1.58, +1.56]	52.69 / 61.35
SaudiTalk long-form (telephone)	133	+10.76 [+9.37, +12.14]	+2.96 [+1.96, +3.96]	51.89 / 63.07

The customer-service sample is the only public set in our domain, and it shows the largest WER gap. It contains a single speaker, however, so its intervals reflect only within-call variation. It should not be read as evidence about call centres in general.

5.3 Adapting the Encoder

Table 5 adds the adapted system. On the validation calls, whose labels are machine transcripts, the best greedy CTC WER rises from 46.87% for the frozen character reader to 49.15% for the sub-word reader with attention adapters after two epochs. It then falls to 40.51% for the hybrid reader, and to 40.01% after the joint stage. Adding the feed-forward and convolution adapters brings it to 38.59%. These stages differ in training length and data, so they form a development history rather than a controlled ablation.

Figure 2 summarises these results.

On the human-labelled sets, the adapted system lowers WER relative to the frozen system on every set, and joint decoding lowers it further on every set where both searches were run. Table 6 compares it with Cohere. On the customer-service call, joint decoding gives 19.20% WER against Cohere's 19.27%, a difference of -0.07 points (interval -3.90 to 3.48), and 0.49 points on its telephone version. The two systems are statistically indistinguishable in WER there. Its CER is significantly lower: 7.77% against 10.05% (difference -2.28, interval -4.45 to -0.21), and 8.39% against 10.92% on telephone audio. On Emirati speech, the WER difference is -1.37 points on clean audio (interval -3.11 to 0.43, not significant) and -3.14 on telephone-coded audio (interval -5.01 to -1.32), where the adapted system is significantly better. Its CER differs by -5.47 and -6.98 points (negative favours the adapted system). On SaudiTalk, Cohere remains ahead in WER by 2.04 points on clean and 4.64 points on telephone-coded audio, both significant, while the adapted system's CER is significantly lower (differences -3.93 and -1.76 points). Joint decoding contributes much of the CER gain. On Emirati speech it lowers CER from 20.77% with CTC search to 14.89%.

Table 7 compares the adapted system with the Nemotron model fine-tuned on SADA. The adapted system has a significantly lower WER on the customer-service call, by -4.92 points

Table 5: Human-labelled test sets, WER / CER (%): Cohere Transcribe Arabic, the frozen system (telephone-SADA reader, word 4-gram) and the adapted system with CTC beam search and a word 4-gram, or with joint CTC/attention search (beam 16) and a sub-word 6-gram; Nemotron is the 0.6B model fine-tuned on SADA. A dash marks a run not made.

Test set	Cohere	Nemotron	Frozen CTC + LM	Adapted CTC + LM	Adapted Joint + LM
Saudi customer-service calls (clean)	19.27 / 10.05	24.12 / 8.58	29.62 / 11.91	20.86 / 10.02	19.20 / 7.77
Saudi customer-service calls (telephone)	20.53 / 10.92	25.41 / 9.85	30.63 / 12.72	22.61 / 10.58	21.02 / 8.39
Casablanca, Emirati (clean)	48.35 / 20.35	53.27 / 17.09	54.63 / 21.26	50.66 / 20.77	46.98 / 14.89
Casablanca, Emirati (telephone)	53.15 / 24.38	56.56 / 19.81	58.82 / 25.38	53.50 / 23.53	50.01 / 17.40
SaudiTalk long-form (clean)	56.75 / 29.51	61.25 / 25.76	65.29 / 29.60	60.29 / 29.78	58.79 / 25.58
SaudiTalk long-form (telephone)	55.80 / 28.40	62.63 / 27.22	66.56 / 31.36	–	60.45 / 26.65

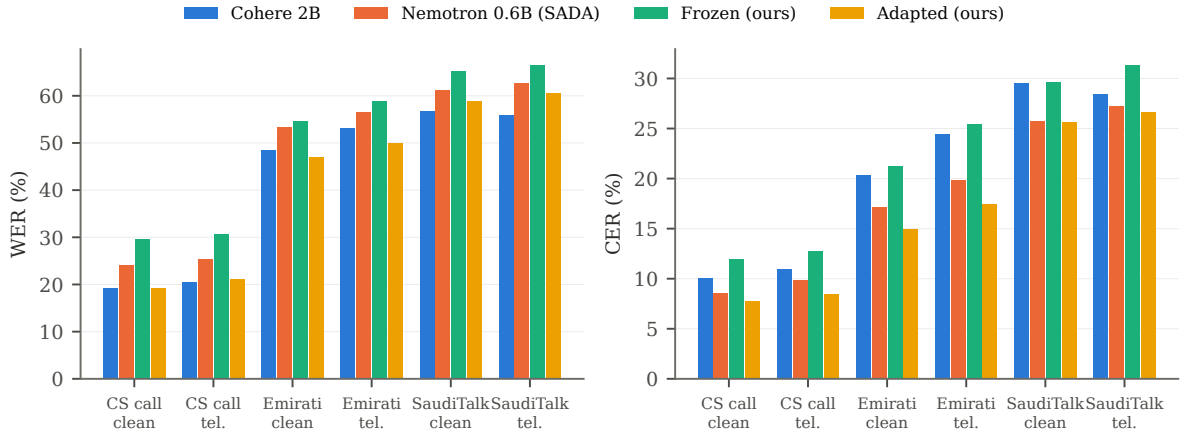

Figure 2: WER and CER on the human-labelled test sets for Cohere, the SADA-fine-tuned Nemotron model and our two systems (adapted: joint search; values in Table 5).

(interval -8.25 to -1.68). It is also significantly lower on Emirati speech, by -6.28 and -6.55 points, and its CER on Emirati speech is lower too. On SaudiTalk its WER is significantly lower, by -2.46 and -2.18 points. The CERs are close, differing by -0.19 points (interval -0.59 to 0.23) on clean audio.

The 16-best lists from the joint search show how much room remains. On the customer-service call, the best of the 16 hypotheses scores 13.06% WER, or 13.84% on telephone audio. This is well below both the top hypothesis and Cohere. On Emirati speech it scores 35.63% and 38.83%. The adapted system already produces many transcripts better than the 2B recogniser's, unlike the frozen system (Section 5.7), so better ranking could still lower its WER. These are oracle bounds, not achieved results.

5.4 Blind Call-Centre Sample

Table 8 adds a third drafting system, the commercial ElevenLabs Scribe v2 [50], on the blind call-centre sample. The adapted system has the lowest WER and CER on clean audio and the lowest CER on telephone-coded audio, where Cohere's WER is slightly lower. The sample is small, however: in paired comparisons on clips drafted by neither system, the WER difference (adapted minus other; negative favours the adapted system) is -5.47 points against Cohere (interval -12.45 to 1.69, 31 clips) and -9.50 points against Scribe (interval -20.10 to 0.25), and among all these comparisons only the CER difference against Scribe on clean audio (-4.41 points, interval -8.58 to -0.87) is statistically significant. The direction agrees with the other call-centre results, but the sample is too small to establish a difference on its own. The annotator checked every clip

Table 6: Paired difference, adapted system minus Cohere, in percentage points, with 95% bootstrap intervals over utterances (10,000 resamples). Negative values favour the adapted system. The last column gives corpus-level WER (Cohere / adapted).

Test set	Search	ΔWER	ΔCER	Corpus WER
Saudi customer-service calls (clean)	Joint + LM	-0.07 [-3.90, +3.48]	-2.28 [-4.45, -0.21]	20.77 / 19.81
Saudi customer-service calls (telephone)	Joint + LM	+0.49 [-2.51, +3.46]	-2.53 [-4.23, -0.84]	21.87 / 21.60
Casablanca, Emirati (clean)	Joint + LM	-1.37 [-3.11, +0.43]	-5.47 [-6.88, -4.10]	47.52 / 47.38
Casablanca, Emirati (telephone)	Joint + LM	-3.14 [-5.01, -1.32]	-6.98 [-8.63, -5.40]	51.75 / 50.06
SaudiTalk long-form (clean)	Joint + LM	+2.04 [+0.64, +3.41]	-3.93 [-5.59, -2.49]	52.69 / 54.34
SaudiTalk long-form (telephone)	Joint + LM	+4.64 [+3.41, +5.86]	-1.76 [-2.85, -0.73]	51.89 / 56.20

Table 7: Paired difference, adapted system minus Nemotron 0.6B (fine-tuned on SADA), in percentage points, with 95% bootstrap intervals over utterances. Negative values favour the adapted system.

Test set	Search	ΔWER	ΔCER
Saudi customer-service calls (clean)	Joint + LM	-4.92 [-8.25, -1.68]	-0.81 [-2.05, +0.38]
Saudi customer-service calls (telephone)	Joint + LM	-4.39 [-7.81, -1.07]	-1.46 [-2.88, -0.12]
Casablanca, Emirati (clean)	Joint + LM	-6.28 [-7.70, -4.88]	-2.20 [-2.92, -1.50]
Casablanca, Emirati (telephone)	Joint + LM	-6.55 [-8.00, -5.13]	-2.41 [-3.17, -1.63]
SaudiTalk long-form (clean)	Joint + LM	-2.46 [-3.35, -1.59]	-0.19 [-0.59, +0.23]
SaudiTalk long-form (telephone)	Joint + LM	-2.18 [-3.02, -1.33]	-0.58 [-1.00, -0.15]

against the audio; the annotator accepted unchanged 23 of 31 Scribe drafts, 18 of 31 drafts from our system and 16 of 30 Cohere drafts, after normalisation). Scribe's median clip WER (25.0%) is no higher than the others' (27.3% for ours, 25.0% for Cohere), but it fails completely, with a WER of 100% or more, on 7 clips against 2 and 3, which raises its mean. Excluding each system's own drafts removes the direct advantage of drafting; a second, independent annotator would be needed to rule out any remaining effect of the drafts on the references.

Because all 15 systems can be scored on the clips not drafted by our system, the blind sample also allows a wider comparison with commercial speech-recognition services, accessed through OpenRouter [51] with the language set to Arabic (Table 9). Three commercial APIs score better than the adapted system, significantly so only GPT-Transcribe: GPT-Transcribe [52] is clearly better (11.05 WER points in our disfavour, interval 6.14 to 16.17), and Chirp 3 [53] and MAI-Transcribe-2 [54] are ahead by smaller, non-significant margins. The adapted system ranks 4 of 15 and has the lowest WER of the self-hostable systems tested by point estimate (its gap to Cohere is not significant); it is significantly better than Whisper large-v3, both Voxtral models, Deepgram Nova-3 [55], GPT-4o Transcribe [56] and Fish Audio [57], and not significantly different from Gemini 3.5 Transcribe [58], Grok STT [59], AssemblyAI [60], Cohere and Scribe. These comparisons rest on 61 clips and one annotator, so they indicate rankings rather than precise gaps.

Figure 3 shows the ranking.

Figure 4 shows the outputs of several systems for one clip. The clip was chosen by a fixed rule rather than by hand: among clips drafted by Scribe (which is therefore not shown) with 7 to 14 reference words, it is the one whose mean WER across the shown systems is closest to the median. Most systems differ only in one or two words, typically a dialect form or a spelling of the same word.

Table 10 brings accuracy, deployment and cost together. The commercial systems that are more accurate cost between $0.10 and $0.96 per audio hour, and they send audio to a third party. The adapted system runs on one GPU at about 139 times real time. On the hardware we used, thousand hours of audio take about 7.2 GPU-hours to process.

Table 8: Public Saudi call-centre dialogue sample with human-corrected references and a balanced blind prefill: each clip's draft came from one of the three systems, hidden from the annotator, and each system is scored only on clips it did not draft (number of clips in brackets). WER / CER (%).

System	Clean	Telephone
Adapted (ours)	29.89 / 11.30 (61)	31.51 / 11.92 (61)
Cohere Transcribe Arabic 2B	34.17 / 14.59 (62)	30.87 / 14.10 (62)
ElevenLabs Scribe v2	35.32 / 17.16 (61)	37.00 / 19.30 (61)

Table 9: Blind call-centre sample: 15 recognisers sorted by clean WER. Each system is scored on the clips that neither it nor our system drafted (n); commercial APIs drafted none. The last column is the paired WER difference, ours minus that system, on clean audio over clips drafted by neither, with a 95% bootstrap interval; * marks intervals that exclude zero. WER / CER (%).

System	Self-hostable	n	Clean	Telephone	ΔWER vs ours
GPT-Transcribe [52]	no	61	18.84 / 8.14	19.21 / 9.13	+11.05 [+6.14, +16.17]*
Chirp 3 [53]	no	61	26.50 / 9.78	25.70 / 10.12	+3.39 [-1.48, +8.17]
MAI-Transcribe-2 [54]	no	61	27.28 / 9.12	24.32 / 9.11	+2.61 [-1.88, +6.63]
Adapted (ours)	yes	61	29.89 / 11.30	31.51 / 11.92	–
Grok STT 1.0 [59]	no	61	31.23 / 17.12	31.94 / 14.25	-1.34 [-8.10, +4.94]
Gemini 3.5 Transcribe [58]	no	61	31.33 / 13.05	29.73 / 11.62	-1.44 [-8.09, +4.66]
AssemblyAI Universal-3.5 Pro [60]	no	61	32.15 / 12.83	32.80 / 13.89	-2.26 [-9.31, +4.17]
Cohere Transcribe Arabic 2B [25]	yes	31	35.06 / 12.11	33.07 / 14.63	-5.47 [-12.45, +1.69]
Deepgram Nova-3 [55]	no	61	36.03 / 13.45	36.97 / 16.22	-6.14 [-12.61, -0.14]*
Voxtral Mini Transcribe [61]	no	61	38.29 / 16.64	39.16 / 17.91	-8.40 [-15.54, -2.17]*
GPT-4o Transcribe [56]	no	61	38.65 / 27.63	44.83 / 30.54	-8.76 [-15.98, -1.83]*
Fish Audio Transcribe 1 Pro [57]	no	61	39.50 / 21.11	38.61 / 21.09	-9.61 [-19.27, -0.91]*
ElevenLabs Scribe v2 [50]	no	30	39.71 / 15.02	39.38 / 17.39	-9.50 [-20.10, +0.25]
Voxtral Small 24B [61]	yes	61	39.74 / 16.71	44.86 / 20.53	-9.85 [-18.27, -2.23]*
Whisper large-v3 [3]	yes	61	44.35 / 22.84	46.74 / 22.57	-14.46 [-22.21, -7.08]*

5.5 Call-Centre Recordings

Table 11 gives results on the in-house calls. With the language model, the telephone-SADA reader reaches 47.20% WER and 25.32% CER. The fine-tuned Nemotron model reaches 46.71% WER and a clearly better 21.47% CER. The WER difference, 0.49 points, is not significant (window bootstrap -2.58 to 3.71; call bootstrap -2.49 to 3.68), but a non-significant difference on 72 windows does not establish equivalence, and the two systems were trained on different data: our reader saw in-domain calls, while Nemotron saw only SADA. Cohere scores 37.62%, a difference of 9.59 points (window bootstrap 6.65 to 12.98; call bootstrap 6.68 to 12.88). Corpus-level WER is 49.14% for our reader, 48.33% for Nemotron and 40.13% for Cohere. The zero-shot Nemotron model scores 51.33%. The adapted system reaches 42.63% WER with CTC search and 42.91% WER (22.47% CER) with joint search.

Table 12 shows why the Cohere comparison on this set must be read with care. On the 30 windows whose reference is an unchanged Cohere draft, Cohere leads the frozen-encoder system by 14.78 points. On the 33 windows whose reference is an unchanged draft from the commercial cloud recogniser, the lead shrinks to 5.40 points (30.88% against 36.28%). Part of Cohere's advantage on the full set is therefore an artefact of how the references were made. The subset drafted by the commercial cloud recogniser is not neutral either, since that system also labelled our training data.

Table 13 separates conversational windows from windows dominated by spoken numbers. For every system, WER on the two kinds differs by at most about two points. Spoken digit strings

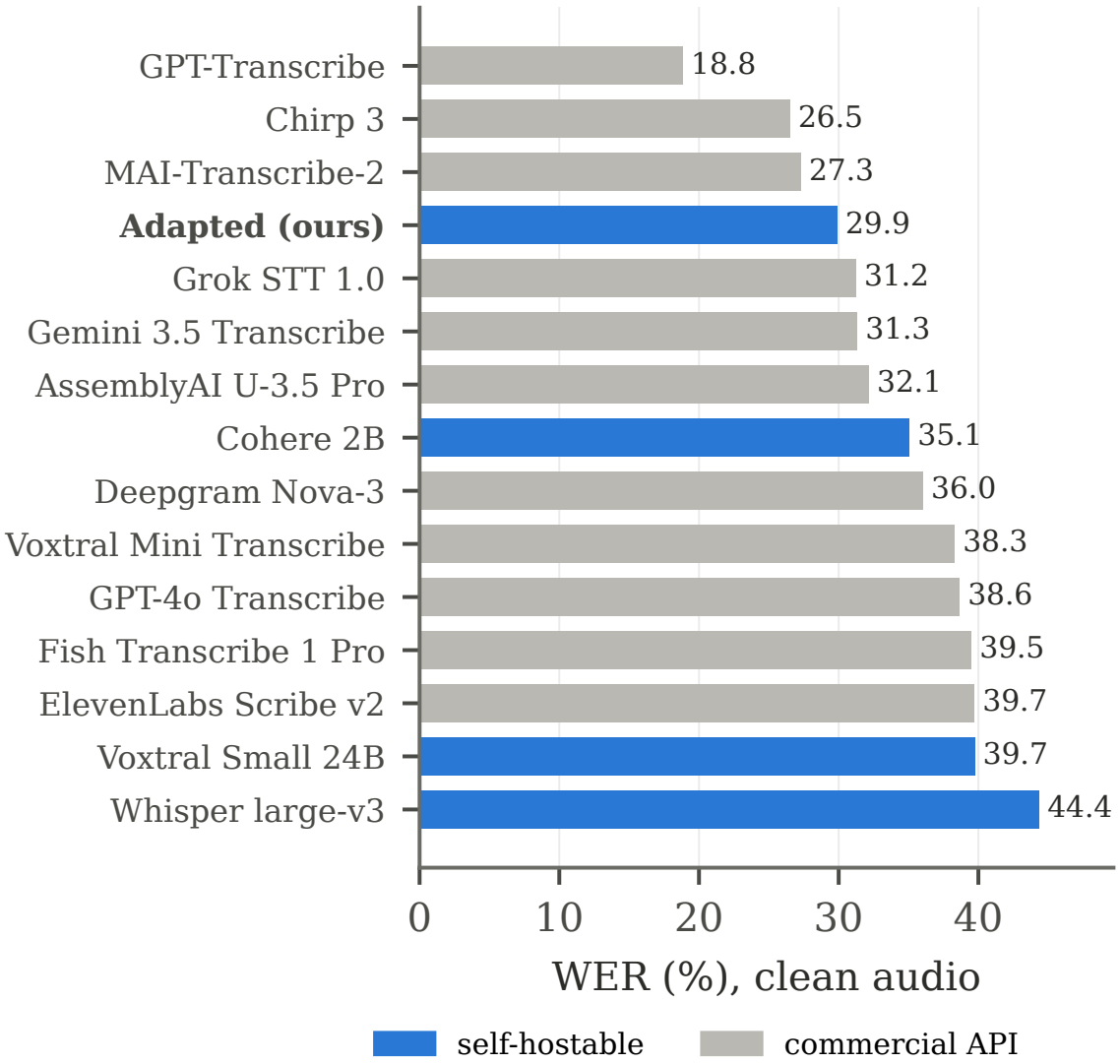

Figure 3: Clean-audio WER of the fifteen systems on the blind call-centre sample, sorted; self-hostable systems in blue. Values and intervals in Table 9.

are therefore not a particular weakness of the frozen encoder relative to the other systems.

5.6 Which Layers Does the Reader Use?

Figure 5 plots the weights. The learned weights of Eq. (1) for the telephone-SADA reader are 25.6% for layer 6, 7.8% for layer 8, 21.8% for layer 10, 25.9% for layer 11, 9.5% for layer 12, and 9.4% for the projected output. Layers 6, 10 and 11 together receive more than two thirds of the weight. These weights should not be read as layer importance. Each layer has its own normalisation and projection, and these can absorb a scale factor. Only the six listed representations were offered to the mix, so a single-layer probe of all twelve layers is needed before any conclusion is drawn about where the encoder holds transcription-relevant detail.

System	WER	Transcript (normalised)
Reference		ابغي هذا ارخص شي يعني ما ابغي تكلف لاني بعدني طالبه
Ours (adapted)	18%	ابو هذا ارخص شي يعني ما ما ابغي تكلف لاني بعدني طالبه
GPT-Transcribe	9%	ابغي هذا ارخص شي يعني ما ابغي تكلف لان بعدني طالبه
Chirp 3	9%	ابغي هذا ارخص شي يعني ما ابغي اتكلف لاني بعدني طالبه
MAI-Transcribe-2	18%	ابغي هذا ارخص شي يعني ما ما ابغي اتكلف لاني بعدني طالبه
Cohere 2B	27%	ابغي هذا ارخص شي يعني ما ما ابغي اتكلف لان بعدني طالبه
Deepgram Nova-3	18%	هذا ارخص شي يعني ما ابغي اتكلف لاني بعدني طالبه [–]
Whisper large-v3	27%	ابغي هذا رخص شي يعني ما بغي تكلف لني بعدني طالبه

Figure 4: Outputs of seven systems for one clip of the blind call-centre sample, after the scoring normalisation. Red words differ from the reference; a grey placeholder marks a missing reference word.

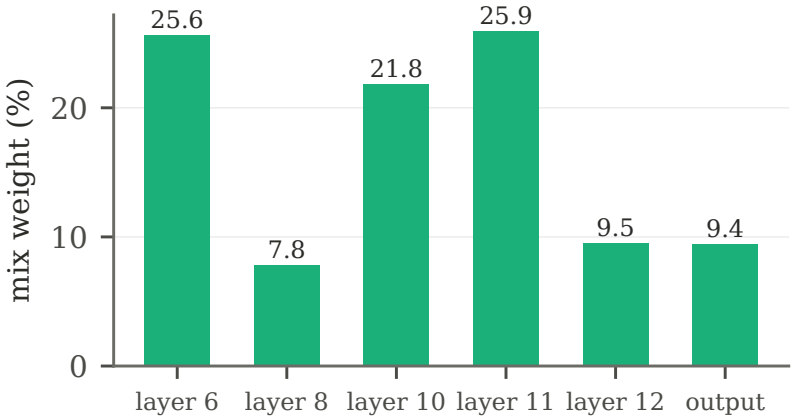

Figure 5: Learned layer-mix weights of the frozen telephone-SADA reader.

5.7 Search or Model?

Could better re-ranking close the call-centre gap? We computed the oracle WER of a wider beam (128 hypotheses, same language model), that is, the lowest WER among the top N hypotheses of each window. The top hypothesis of this wider search scores 47.63%, slightly different from the 47.20% of the beam-64 system. The oracle falls to 46.03% for the top 8 and 45.10% for the top 128, and pooling the beams with and without the language model gives 44.92%. Even a perfect re-ranker would stay above Cohere's 37.62%. The same holds on the 33 windows with cloud-recogniser references, which Cohere did not draft: there the top-128 oracle is 34.97%, against 30.88% for Cohere. The gap therefore lies in the acoustic model and its candidates, not in search.

5.8 Speed and Compute Cost

Table 14 reports throughput measured on identical audio, one system at a time on an otherwise idle GPU, with batches sorted by length so that no system is penalised by padding; on the GPU both systems use batches of 16, and Cohere is also shown with 32. All systems run in PyTorch without further inference optimisation. The encoder on its own processes audio at 348 times real time on the GPU; the complete pipeline with greedy decoding, including feature extraction

Table 10: Final comparison on the blind call-centre sample (clean audio, clips drafted by neither the system nor ours). Speed is end-to-end throughput measured on one Quadro RTX 6000 (Table 15); cloud APIs were not timed comparably. Price is the list price for batch transcription per audio hour, fetched on 2026-10-16 from each vendor's pricing page (MAI-Transcribe-2 via OpenRouter, for which no vendor price is published); for open-weight models we accessed through an API, the API price is shown. Self-hosted systems pay only for their own hardware.

System	Self-hostable	WER / CER (%)	Speed	Price per hour
GPT-Transcribe	no	18.84 / 8.14	cloud API	$0.27
Chirp 3	no	26.50 / 9.78	cloud API	$0.96
MAI-Transcribe-2	no	27.28 / 9.12	cloud API	$0.10
Adapted (ours)	yes	29.89 / 11.30	139×	self-hosted
Grok STT 1.0	no	31.23 / 17.12	cloud API	$0.10
Gemini 3.5 Transcribe	no	31.33 / 13.05	cloud API	$0.30
AssemblyAI Universal-3.5 Pro	no	32.15 / 12.83	cloud API	$0.21
Cohere Transcribe Arabic 2B	yes	35.06 / 12.11	80×	self-hosted
Deepgram Nova-3	no	36.03 / 13.45	cloud API	$0.26
Voxtral Mini Transcribe	no	38.29 / 16.64	cloud API	$0.18
GPT-4o Transcribe	no	38.65 / 27.63	cloud API	$0.36
Fish Audio Transcribe 1 Pro	no	39.50 / 21.11	cloud API	$0.36
ElevenLabs Scribe v2	no	39.71 / 15.02	cloud API	$0.22
Voxtral Small 24B	yes	39.74 / 16.71	not measured	$0.24
Whisper large-v3	yes	44.35 / 22.84	not measured	$0.03

and the reader, runs at 257 times real time and needs 1.49 GiB of GPU memory. At the same batch size, Cohere Transcribe Arabic runs at 157 times real time with 4.76 GiB, so the frozen-encoder system is 1.6 times faster and uses 3.2 times less memory. The speed advantage on the GPU is modest, however: with a larger batch Cohere reaches 222 times real time, because its autoregressive decoder produces few tokens for short utterances and parallelises well.

On CPU, often the deployment target in call centres, the difference is larger. With 16 threads, the frozen-encoder system runs at 31.10 times real time with greedy decoding, which is 3.4 times faster than Cohere (9.20) and 1.4 times faster than the 0.6B Nemotron model with greedy decoding (22.60). The accuracy figures above rely on 4-gram beam search, which adds work on the CPU: one process decodes at 84 times real time, or about 12 core-hours per 1,000 hours of audio, though utterances can be decoded in parallel. Nemotron's batched GPU inference failed with a memory-access error on the audio of Table 14, so only its CPU figures appear there; it was timed on the GPU on different audio in Section 5.9. These measurements cover 13 minutes of short utterances on CPU and one GPU type. They describe the frozen system with greedy decoding only. The adapted system is timed in Section 5.9.

5.9 Fast Joint Decoding

Joint CTC/attention search is what gives the adapted system its accuracy, but our first implementation ran at only 1.7 times real time, because every search step issued many small GPU operations and scored CTC prefixes frame by frame in Python. We rewrote it in six steps, checking after each step how many transcripts changed: CTC prefix scores are computed by a compiled routine; the language model is queried only for the candidates that survive the attention pre-beam; the attention decoder keeps a key/value cache and projects the encoder memory once per utterance; one decoder call serves all utterances of a batch at every step; the decoder runs in 16-bit floating point, which the GPU executes natively, instead of bfloat16, which it emulates; and the encoder processes length-sorted batches rather than one segment at a time. Batching the search across utterances left all 60 checked transcripts unchanged, and half

Table 11: In-house call-centre recordings (72 windows, 8 kHz; machine-drafted references reviewed by one annotator, see Section 4): mean per-window WER and CER (%) with greedy CTC decoding and with the 4-gram language model. The reference systems use their own decoders, so their scores appear in the right-hand columns.

	Greedy		+ 4-gram LM	
System	WER	CER	WER	CER
Calls only	60.43	27.10	49.00	25.99
+ SADA, telephone-coded	58.69	25.61	47.20	25.32
+ SADA, clean	58.15	25.65	47.26	25.47
Nemotron 3.5 ASR 0.6B, zero-shot (beam 8)	–		51.33	25.83
Nemotron 3.5 ASR 0.6B, fine-tuned on SADA (beam 8)	–		46.71	21.47
Cohere Transcribe Arabic 2B (greedy)	–		37.62	19.67

Table 12: Call-centre WER (%) split by where the reference came from: Cohere draft accepted unchanged (30 windows), commercial cloud recogniser's draft accepted unchanged (33), and edited by the annotator (9).

System	Cohere draft	Cloud ASR draft	Edited
Calls only	60.11	37.87	52.75
+ SADA, telephone-coded	58.38	36.28	50.01
+ SADA, clean	57.54	37.14	50.11
Nemotron 0.6B, fine-tuned	56.16	35.94	54.72
Cohere 2B	43.60	30.88	42.37

precision changed two of 198.

Table 15 gives end-to-end throughput, from audio to text, on one idle GPU. With one process the optimised pipeline runs at 68 times real time, and with four processes sharing the GPU at 139 times, against 1.7 times for the original search on different audio; WER on the telephone customer-service call moves from 20.32% to 20.48%. For comparison, CTC search with the word language model runs at 136 times real time, Cohere at 80 times with batches of 16 (higher in Table 14, which used shorter audio), and Nemotron at 183 times. The joint search is therefore no longer the slow path. The two halves of the table were measured on different audio, so they are not directly comparable; on that basis the adapted system's 139-fold throughput is likely, but not shown, to exceed Cohere's on this GPU. The optimised pipeline differs slightly from the search used for the accuracy tables, which ran one utterance at a time in bfloat16 with a beam of 16; batching and half precision change a few transcripts, which is why WER differs slightly between Table 5 and Table 15. Merging the adapters into the encoder weights changed a few transcripts for little speed gain, so we kept them separate.

6 Discussion

What the frozen encoder can do The speech encoder that Gemma 4 and EmbeddingGemma 2 share stays frozen and is never adapted to Arabic transcription. A small reader on top of it reaches a CER statistically indistinguishable from that of a 2B dedicated recogniser on five of the six human-labelled conditions we tested (Table 4). On real call-centre audio, its WER is not significantly different from that of a 0.6B recogniser fine-tuned on SADA, although its CER is clearly worse. Only 22.6M parameters are trained, so adapting to a new domain is cheap. A deployment that already runs Gemma 4 or EmbeddingGemma 2 can add transcription without loading a second large encoder. The system is also cheaper to run. It is about 3.4 times faster than the 2B recogniser on CPU, with 3.2 times less GPU memory.

Table 13: Call-centre WER (%) by window type: conversational speech (43 windows) and windows dominated by spoken numbers such as identity, phone and account numbers (29).

System	Speech	Numbers
Calls only	49.25	48.63
+ SADA, telephone-coded	47.70	46.47
+ SADA, clean	47.16	47.41
Nemotron 0.6B, fine-tuned	47.41	45.67
Cohere 2B	37.36	38.00

Table 14: Speed and compute cost on identical audio with length-sorted batches. RTFx is seconds of audio processed per second of wall time, including feature extraction and greedy decoding. GPU: one NVIDIA Quadro RTX 6000 (24 GB), 56 minutes of audio (Casablanca Emirati test). CPU: 16 threads of an Intel Xeon Gold 6230, 13 minutes of audio, batch 8, 32-bit floats. Cost is given as GPU-hours or CPU core-hours per 1,000 hours of audio. The 4-gram beam search adds 12 CPU core-hours per 1,000 hours (one process, RTFx 84).

		GPU			CPU (16 threads)	
System	Params (M)	RTFx	Mem. (GiB)	GPU-h	RTFx	core-h
Frozen encoder + reader (greedy)	328	257	1.5	3.9	31.1	514
encoder only	305	348	–	2.9	34.4	–
Cohere Transcribe Arabic (greedy), batch 16	2,066	157	4.8	6.4	9.2	1,739
Cohere Transcribe Arabic (greedy), batch 32	2,066	222	5.2	4.5	–	–
Nemotron 0.6B fine-tuned (greedy)	638	–	–	–	22.6	708
Nemotron 0.6B fine-tuned (beam 8)	638	–	–	–	19.4	825

What adaptation adds Adding 3.1M adapter parameters, sub-word units, an attention decoder and joint decoding makes the system statistically indistinguishable from the 2B recogniser in WER on the customer-service call and on clean Emirati speech. On telephone-coded Emirati speech it is ahead of that recogniser. It also yields a significantly lower CER on all six conditions of the further human-labelled sets. Even the frozen system's 128-best oracle stayed above Cohere's WER on the call-centre set. The adapted system's 16-best lists, by contrast, contain many such transcripts, so better rescoring is a natural next step.

What it cannot yet do On long-form SaudiTalk speech, the adapted system still trails the dedicated recogniser in WER, although its CER there is lower. Our readers are bidirectional. Unlike Nemotron, neither system streams, even though the encoder is causal.

Limitations Every configuration was trained once, so we have no estimate of seed variance. The adapted system's training and language-model data also differ from the frozen system's, which means its gains cannot be attributed to the encoder adapters alone. Its decoding weights were tuned at an earlier stage on validation calls whose machine labels favour CTC over joint search, the opposite of every human-labelled set; a validation set with human labels would be a better guide. We did not compare against other frozen encoders, such as the Whisper or w2v-BERT encoders, using the same reader, nor against the full Gemma 4 model used as a recogniser. Such controls would show how much of the result is specific to this encoder. The call-centre training labels are machine transcripts from two systems, one of which is the strongest baseline, so the reader is partly distilled from it. The call-centre references are mostly unedited machine drafts, as Section 4 explains, and the windows were selected to be hard. The blind call-centre sample was corrected by one annotator, the author, and is small. The only public customer-service set with original human references is a single call, so the adapted system's results there rest on one speaker. The adapted system also trains on voicebot calls with machine labels.

Table 15: Throughput of the adapted system and the reference recognisers on one idle Quadro RTX 6000 (RTFx = seconds of audio per second of wall time, end to end). Upper part: the optimised joint CTC/attention pipeline with the sub-word LM (827 s of audio, batches of 16 utterances) and its WER/CER on the telephone customer-service call. Lower part: the original joint search and the other systems on the 1673 s of the in-house call windows.

System	RTFx	CS-call tel. WER / CER
Optimised joint search, 1 process	68	20.32 / 8.34
Optimised joint search, 2 processes	106	20.48 / 8.38
Optimised joint search, 3 processes	133	20.48 / 8.38
Optimised joint search, 4 processes	139	20.48 / 8.38
Original joint search (beam 10)	1.7	
Adapted, CTC greedy (batch 1 / 16)	174 / 141	
Adapted, CTC + word LM (12 CPU processes)	136	
Cohere Transcribe Arabic 2B (batch 1 / 16)	15 / 80	
Nemotron 0.6B (batch 1 / 16)	131 / 183	

Telephone conditions are simulated with G.711 μ-law only, whereas real networks add other codecs, packet loss and echo. The SADA and Casablanca licences forbid commercial use, so the SADA-trained readers and language model are research artefacts only. Speed was measured on one GPU type and one CPU type, and only our own system received inference optimisation; Cohere and Nemotron were run with their standard implementations.

7 Conclusion

To our knowledge, this is the first Arabic speech-recognition study of the audio encoder shared by Gemma 4 and EmbeddingGemma 2, which we verified to be identical in both releases except for the final projection into the text model. Kept frozen and read by a 22.6M-parameter CTC model, the encoder supports Saudi and Gulf recognition whose CER is statistically indistinguishable from that of a 2B dedicated recogniser on five of six human-labelled conditions. It does so at several times the CPU speed and about a third of the GPU memory. With 3.1M adapter parameters, a hybrid reader and joint decoding, and still without changing a single encoder weight, the system is statistically indistinguishable from the 2B recogniser in WER on a public Saudi customer-service call and on clean Emirati speech. It is significantly better on telephone-coded Emirati speech and has a significantly lower CER on all six conditions of the further human-labelled sets. In WER, it still trails on long-form Saudi speech. It is also significantly better than a 0.6B recogniser fine-tuned on SADA in WER on all six of those conditions. An optimised joint decoder runs it at about 139 times real time on one GPU. On a blind, human-corrected public call-centre sample it has the lowest WER of the self-hostable systems among the 15 tested, while three commercial APIs score better, significantly so only GPT-Transcribe. We further show that references built by correcting machine drafts can inflate the apparent quality of the drafting system. We recommend that such references be reported by origin.

Ethics and Data Statement

The in-house call recordings were processed for this research and are not released; no transcript or audio excerpt is reproduced in this paper, and all results are aggregate scores. The recordings were used under an agreement with the data owner for research. SADA (CC BY-NC-SA), Casablanca (CC BY-NC-ND), MASC (CC BY 4.0), SaudiTalk and the two public call-centre samples were used for research only, under their published terms; no model trained on non-commercial data is distributed for commercial use.

Code and Data Availability

Code for both systems, including the fast joint decoder, and every result on public test sets are available at `https://github.com/AmrKhalil9/frozen-embeddinggemma2-arabic-asr`, together with the sourced list prices. The in-house call recordings, their transcripts, the results on them and the trained weights are not released.

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Competing Interests

The author declares no competing interests.

Use of AI Assistance

During this work the author used Claude (Anthropic) to help write code and to draft and edit the manuscript, and used other AI assistants to review the experimental plan. The author reviewed and verified all content and takes full responsibility for it. Every number in the tables and text is generated by a script directly from the released result files.

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